



Operations Research

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Exercise Sheet 3

Training Exercises

Exercise T3.1. Revised Simplex

$$\begin{array}{ll} \text{Max} & 2x_1 + x_2 + 3x_3 + x_4 + 2x_5 \\ \text{s.t.} & \end{array}$$

Solve the following linear optimization problem using the revised simplex method.

$$\begin{array}{l} x_1 + 2x_2 + x_3 + x_5 \leq 100 \\ x_2 + x_3 + x_4 + x_5 \leq 80 \\ x_1 + x_3 + x_4 \leq 50 \\ x_1, \dots, x_5 \geq 0 \end{array}$$

Exercise T3.2. Lemonade Production

The lemonade factory *LiMo-United (LMU)* produces two different lemonades: *AquaMarin* and *BubbleBurst*. Both lemonades use *caffeine*, *oranges* and *sugar* as raw materials. The following table lists the available capacities of the raw materials in kilograms as well as the required quantities of the raw materials (also in kilograms) to produce 10 liters of Aquamarine or BubbleBurst.

Raw material	AquaMarin	BubbleBurst	Capacity
Caffeine	1	2	20
Oranges	2	2	30
Sugar	1	2	25
Sales price per 10 liters	12€	18€	
Costs per 10 liters	10€	15€	

Employee Mattea has already modeled the above production problem as a linear program (P) and determined the optimal table using the simplex algorithm.

$$\begin{array}{ll} (P) \text{ Max} & 2x_1 + 3x_2 \\ \text{s.t.} & \\ & x_1 + 2x_2 + x_3 = 20 \\ & 2x_1 + 2x_2 + x_4 = 30 \\ & x_1 + 2x_2 + x_5 = 25 \\ & x \in \mathbb{R}_{\geq 0}^5 \end{array}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	0	0	1	0.5	0	35
x_2	0	1	1	-0.5	0	5
x_1	1	0	-1	1	0	10
x_5	0	0	-1	0	1	5

For the following subtasks it is assumed that LMU always acts in a profit-maximizing manner.

- a) Explain the meaning of the objective function, the constraints and the decision variables of (P) .
- b) How many liters of AquaMarin and BubbleBurst must be produced according to the optimal production plan?
- c) LMU has the option to buy another 1kg of oranges. What is the maximum price that LMU should pay for this?
- d) Another company was unable to obtain enough sugar due to supply shortages. It therefore proposes to buy 4kg of sugar from LMU for a total of 7€. Should LMU accept this proposal?
- e) Calculate the upper and lower limits for the sales price of 10L AquaMarin within which the optimal basis does not change. Briefly explain (without values) how the optimal production plan and resource stocks change if the sales price exceeds the upper limit.
- f) Due to supply shortages, the availability of oranges is unfortunately significantly worse this year, which is why LMU is considering reducing the stock of oranges in the future.
 - (i) What is the minimum number of kilograms of oranges required to continue to make it economically viable to produce both AquaMarin and BubbleBurst?
 - (ii) How do the shadow prices and reduced costs change if the amount of oranges falls below this value?
- g) Mattea proposes to produce a new drink. To produce 10 liters of *CitroCider* you need 1kg of caffeine, 1kg of oranges and 3kg of sugar. If the production costs per 10 liters are 8€ how high must the selling price of 10 liters of CitroCider be set for production to make economic sense? *Note: You can assume that all shadow prices are still valid for deviations of up to 1kg.*

Exercise T3.3. Sensitivity

Consider the following linear program and the corresponding optimal tableau:

$$\begin{aligned}
 (P) \quad & \text{Max } -2x_1 + 3x_2 \\
 \text{s.t.} \quad & x_1 + 3x_2 \leq 7 \\
 & x_1 + \frac{1}{2}x_2 \geq 1 \\
 & 2x_1 + 2x_2 \leq 4 \\
 & x \geq 0
 \end{aligned}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
Z	5	0	0	0	$3/2$	6
x_3	-2	0	1	0	$-3/2$	1
x_2	1	1	0	0	$1/2$	2
x_4	$-1/2$	0	0	1	$1/4$	0

- a) Convert (P) to standard form (P_{SF}) . Is (P_{SF}) already in canonical form? Explain.
- b) Read an optimal solution of (P) from the given tableau.
- c) Is the solution of (P) unique? Explain.
- d) In which range can the objective function coefficient of the variable x_1 fluctuate without the optimal basis changing?
- e) In which range can the objective function coefficient of the variable x_2 fluctuate without the optimal basis changing?

f) Determine the limits of all constraints in which the optimal basis does not change.

Self-study Exercises

Exercise S3.1. Modeling

Reformulate the following optimization problem into a linear optimization problem in normal form.

$$\begin{aligned}
 & \text{Min}_{\text{s.t.}} \max\{x_1, x_4\} \\
 & | -x_1 + x_2 + x_3 + x_4 | \leq 10 \\
 & \frac{x_2 - x_4}{x_1 + x_3 + 1} \leq 4 \\
 & \max\{x_1, x_2\} \leq \min\{x_3, x_4\} \\
 & x_1, x_2, x_3, x_4 \geq 0
 \end{aligned}$$

Exercise S3.2. Revised Simplex

Consider the following linear optimization problem:

$$\begin{aligned}
 & \text{Max}_{\text{s.t.}} 4x_1 + 2x_2 + 3x_3 + 5x_4 \\
 & x_1 + x_3 + x_5 = 12 \\
 & 3x_1 + x_2 + 2x_3 + x_4 + x_6 = 24 \\
 & 2x_1 + 2x_2 + 3x_3 + 2x_4 + x_7 = 18 \\
 & x \geq 0
 \end{aligned}$$

Perform one step of the revised simplex algorithm, starting from the initial basis $\{x_4, x_5, x_6\}$. Interpret the result. Are there any further steps necessary?

Exercise S3.3. Simplex - Matrix notation

Below, you are given a general LP with parameters A , an $m \times n$ matrix, c and b , vectors of sizes n and m . Note that $\text{rank}(A) = \text{rank}(A \mid b) = m < n$.

$$\begin{aligned}
 & \max_{x \in \mathbb{R}^n} c^T x \\
 & \text{s.t. } Ax = b \\
 & x \geq 0.
 \end{aligned}$$

The following simplex tableau solves the above LP, where I is the identity matrix of size $m \times m$.

Base	Coefficients		Res.
	x_B	x_N	
Z	0	$-(c_N - N'^T c_B)^T$	$c_B^T b'$
x_B	I	$B^{-1}N$	$B^{-1}b$

- Explain every symbol that appears in the tableau ($x_B, x_N, Z, c_N, N', c_B, b', B, N, b$): what it stands for, how it is written in terms of LP components (c, x, A, b) , whether it is a vector or a matrix, and its size.
- What do $B^{-1}b, c_B^T b'$, and $-(c_N - N'^T c_B)^T$ stand for in the optimal simplex tableau?

- c) Give some numerical values to the LP parameters c, A, b so that we have a problem with 2 variables that would require the two-phase method (or big M) for the simplex algorithm. Make sure that your LP is not unbounded.

Answer the following questions according to below optimal simplex tableau.

Base	Coefficients				Res.
	x_1	x_2	x_3	x_4	
Z	0	0	2	2	57
x_1	1	0	$2/3$	$-1/6$	2
x_2	0	1	$-1/3$	$1/3$	5

- d) Find the corresponding LP parameter, A matrix, for the above optimal simplex tableau. Note that N (the constraint coefficient matrix of x_3, x_4) is the identity matrix of size 2×2 .
- e) Find the corresponding LP parameter, b vector, for the above optimal simplex tableau.
- f) Find the corresponding LP parameter, c vector, for the above optimal simplex tableau. Note that c_1 (the objective coefficient of x_1) is equal to 6.

Exercise S3.4.

$$\begin{array}{ll} \text{Max} & 3x_1 + 5x_2 \\ \text{s.t.} & \end{array}$$

Solve the following linear optimization problem using the revised simplex algorithm (in matrix notation):

$$\begin{array}{ll} & x_1 \leq 4 \\ & 2x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0 \end{array}$$

Exercise S3.5. PopCo

The snack manufacturer PopCo draws up its production plan for chocolate bars in November 2019 to maximize potential profit. It has 3 chocolate bars in its range: CrackerJoe, CrispMoe and CookieDoe. CrackerJoe requires 1 unit of each walnut, peanut and chocolate. CrispMoe requires 1 peanut and 2 chocolate. CookieDoe requires 2 walnut and 1 chocolate. This consumption, the available capacity of the materials for November and the unit price of each bar are shown in the table below. Due to machine constraints, no more than 100 pieces of CookieDoe can be produced.

Chocolate bar	CrackerJoe	CrispMoe	CookieDoe	capacity in E
Walnut	1		2	300
Peanut	1	1		400
Chocolate	1	2	1	1000
Unit price in EUR	2	3	4	

- a) Set up the corresponding linear program.

Let x_1, x_2, x_3 be the variables for the quantity produced for CrackerJoe, CrispMoe and CookieDoe; s_1, s_2, s_3 are the slack variables for walnut, peanut and chocolate; s_4 is the slack variable for the CookieDoe constraint. The optimal table looks like this:

Base	Coefficients							Res.
	x_1	x_2	x_3	s_1	s_2	s_3	s_4	
z	1	0	0		3	0	4	
	1	0	0		0	0	-2	
	1	1	0		1	0	0	400
	-1	0	0		-2	1	-1	100
	0	0	1		0	0	1	

- b) Complete the optimal tableau.
- c) State the optimal production plan and the achievable profit.
- d) Calculate to what extent the price of CrispMoe can fluctuate without changing the optimal basis.
- e) What is the maximum amount that PopCo should be willing to pay for an additional unit of walnut / peanut / chocolate? Give reasons.
- f) PopCo could buy 51 unit of new peanuts for a total price of 152 EUR. Should PopCo accept the offer? Give reasons in detail. (*Note: Pay attention to the validity ranges of the shadow prices! You can assume without proof that each newly calculated shadow price is valid for deviations of up to 1 unit.*)
- g) A new bar, CrunchyFlow, uses 1 walnut and 2 peanuts. At what minimum selling price would the production of CrunchyFlow be profitable?

Exercise S3.6. Paper mill

Mertens Paper Mill converts various types of pulpwood into low, medium and high quality newsprint. The consumption for each class, the availability of each wood species and the selling price per ton of newsprint are shown in the following table:

Pulpwood	Low	Medium	High	Availability
European beech	2	2	1	180 t
Pine	1	2	3	120 t
Spruce	1	1	2	160 t
Selling price	€ 900	€ 1000	€ 1200	

Let x_1, x_2, x_3 be the variables for the quantity of low, medium and high quality newsprint produced (in t). Employee Georg used the simplex method to determine an optimal production plan. Unfortunately, he wrote so illegibly that some entries can no longer be deciphered.

Base	Coefficients						Res.
	x_1	x_2	x_3	s_1	s_2	s_3	
z	0	200		300	300	0	
	1	0.8		0.6	-0.2	0	84
	0	0.4		-0.2	0.4	0	
	0	-0.6		-0.2	-0.6	1	

- a) Set up the corresponding linear program.
- b) Complete the optimal table above and state the optimal production plan and profit.

- c) What are the shadow prices and within what limits are they valid? Explain their economic significance.
- d) Calculate to what extent the price of low-quality newsprint can fluctuate without the optimal basis changing.
- e) Calculate to what extent the price of medium-quality newsprint can fluctuate without the optimal basis changing.
- f) Mr. Mertens could buy another ton of pine for 250 euros. Should he accept the offer? Explain your decision.
- g) 1t luxury newsprint uses 3t red beech, 3t pine and 3t spruce. Calculate the minimum selling price at which the production of this class of newsprint would be profitable.
- h) Mr. Mertens can exchange European beech for spruce on the market. For 2t spruce you get 1t European beech. How many t spruce should he exchange if the current basis in the optimal table is to be maintained?
- i) To simplify sales, Mr. Mertens is considering only selling newsprint in 5t increments.
 - (i) How does this change the problem in a)?
 - (ii) Is the current optimal solution also optimal for the adapted problem? Give reasons.
 - (iii) What does the current optimal solution tell you about the achievable profit of the adapted problem?